

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2013-14

Paper: BSC-02A — Elements of Geology - I (Ancillary)

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem II/17

*Instructions: Answer **five** questions in all, selecting at least **two** from each of Sections A and B. Section-C is compulsory. All questions carry equal marks.*

SECTION — A

Question 1. Describe the geological work of a river. [14]

Question 2. Write a detailed note on **any two** of the following: [2 × 7 = 14]

- (a) Parts of folds
- (b) Ice ages
- (c) Dip and strike

Question 3. Discuss the relevance and importance of Earth System Science vis-à-vis its different branches. [14]

Question 4. What are sedimentary rocks? Write a detailed note on their classification giving suitable examples. [14]

SECTION — B

Question 5. Explain the mechanism of an earthquake with a neat sketch. Discuss the societal and environmental impact of earthquakes. [14]

Question 6. Write short notes on **any four** of the following: [4 × 3½ = 14]

- (a) Barchans
- (b) Playa
- (c) Chemical weathering
- (d) Moraines
- (e) Nunataks
- (f) Exfoliation

Question 7. How do you distinguish between **any two** of the following mineral pairs based on their diagnostic physical properties? [2 × 7 = 14]

- (a) Quartz and Olivine
- (b) Garnet and Muscovite
- (c) Biotite and Tourmaline
- (d) Plagioclase and Hornblende

SECTION — C

Question 8. Explain the following in 2 to 3 sentences:

[14 × 1 = 14]

- (a) Gabbro
- (b) Asthenosphere
- (c) Plate tectonics
- (d) Normal fault
- (e) Meteorology
- (f) Lustre in minerals
- (g) Mohs' scale
- (h) Richter scale
- (i) Continental drift
- (j) Thrust fault
- (k) Mass extinction
- (l) Denudation
- (m) "Present is the key to the past"
- (n) Marble